

side do not create any problem. Put the inlet and the outlet pipes inside the aluminium water cooling block (see Fig. 7 and Fig. 8) and connect the water pump and thermoelectric wires to a 5V DC power supply.

Assembly

With the cooling water pump reservoir and whole cooling system ready



Fig. 8: Aluminium block for water cooling



Fig. 9: RPi placed in the case



Fig. 10: The reservoir fixed with RPi in the case

to mount, you can now assemble the components and start making a block of the supercomputer. First fix the RPi single-board computer inside the case you have designed, as shown in Fig. 9, and mount the first cooling block on that. Next, mount the second RPi inside the case, making sure the ports are exposed and accessible through the cuts in the case that were made for that purpose.

Fix the second RPi with a screw and then fix the thermoelectric and cooling block on that as well. Then cover top side of the case. To make it look cool and to see the water flowing, use a transparent acrylic sheet to close the case. The details can be seen in Fig. 10 through Fig. 12.

Refer the connection diagram shown in Fig. 13 and accordingly connect the water pump and thermoelectric cooler to 5-12V, 2A DC power supply and power the RPi with 2A, 5V DC power supply.



Fig. 11: Top view



Fig. 12: Fixing of the water cooler module

Software

Install the latest Raspbian OS in the RPi. Create a master unit that will divide the complex task to all its constituent nodes and control them for processing and calculations, like a supercomputer. To do that, open the Linux terminal and set up the MPI. Install MiPIH and prepare the master node that divides and controls the task among the cluster of computers.

```
sudo apt-get update
```

Change the 'pi' user password.

Under 'Networking,' change the hostname to nodeX, replacing X with a unique number (node1, node2, etc). Node1 will be our master unit.

Setup Ethernet:

```
sudo nano /etc/dhcpd.conf
```

Go to the end of file and add the following:

```
interface eth0
static ip_address=10.0.0.1/24
```

Then create unique identity:

```
ssh-keygen -t rsa
Setup master node
ssh-copy-id 10.0.0.1
```

Now the master node is ready. Open the terminal of the second node and connect Ethernet to both the RPi with Ethernet connector and repeat the above steps in that as well. For adding more nodes (which can be in thousands) create the bootable image of the above RPi setup and prepare the SD card of all those nodes with the same image.

Connect all of them with an Ethernet connector and boot all of them and set them as nodes of the master using the following commands

and configurations. Open the terminal and run the following on each node of the supercomputer:

```
sudo apt install mpich
python3-mpi4py
```

Once complete, test MPI is working on each node:

```
mpirun -n 1 hostname
```

From node1 (10.0.0.1), run the following command: